

Query 1 — Industry-Specific Salary Trends

```
from pyspark.sql import SparkSession

# Start a Spark session
spark = SparkSession.builder.config("spark.driver.host", "localhost").appName("JobPostingsAnalysis").getOrCreate()
spark.catalog.clearCache()

# Load the CSV file into a Spark DataFrame
df = spark.read.option("header", "true").option("inferSchema", "true").option("multiLine", "true").option("escape", "\\").csv("data/lightcast_job_postings.csv")

# Register the DataFrame as a temporary SQL view
df.createOrReplaceTempView("job_postings")

# Show Schema and Sample Data
print("----This is Diagnostic check, No need to print it in the final doc----")

# comment the lines below when rendering the submission
df.printSchema()
df.show(5)
```

```
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/27 16:24:29 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform: java classes where applicable
26/05/27 16:24:53 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. Please use writeAsText to view the full text.

----This is Diagnostic check, No need to print it in the final doc----
root
 |-- ID: string (nullable = true)
```

```

|-- LAST_UPDATED_DATE: string (nullable = true)
|-- LAST_UPDATED_TIMESTAMP: timestamp (nullable = true)
|-- DUPLICATES: integer (nullable = true)
|-- POSTED: string (nullable = true)
|-- EXPIRED: string (nullable = true)
|-- DURATION: integer (nullable = true)
|-- SOURCE_TYPES: string (nullable = true)
|-- SOURCES: string (nullable = true)
|-- URL: string (nullable = true)
|-- ACTIVE_URLS: string (nullable = true)
|-- ACTIVE_SOURCES_INFO: string (nullable = true)
|-- TITLE_RAW: string (nullable = true)
|-- BODY: string (nullable = true)
|-- MODELED_EXPIRED: string (nullable = true)
|-- MODELED_DURATION: integer (nullable = true)
|-- COMPANY: integer (nullable = true)
|-- COMPANY_NAME: string (nullable = true)
|-- COMPANY_RAW: string (nullable = true)
|-- COMPANY_IS_STAFFING: boolean (nullable = true)
|-- EDUCATION_LEVELS: string (nullable = true)
|-- EDUCATION_LEVELS_NAME: string (nullable = true)
|-- MIN_EDULEVELS: integer (nullable = true)
|-- MIN_EDULEVELS_NAME: string (nullable = true)
|-- MAX_EDULEVELS: integer (nullable = true)
|-- MAX_EDULEVELS_NAME: string (nullable = true)
|-- EMPLOYMENT_TYPE: integer (nullable = true)
|-- EMPLOYMENT_TYPE_NAME: string (nullable = true)
|-- MIN_YEARS_EXPERIENCE: integer (nullable = true)
|-- MAX_YEARS_EXPERIENCE: integer (nullable = true)
|-- IS_INTERNSHIP: boolean (nullable = true)
|-- SALARY: integer (nullable = true)
|-- REMOTE_TYPE: integer (nullable = true)
|-- REMOTE_TYPE_NAME: string (nullable = true)
|-- ORIGINAL_PAY_PERIOD: string (nullable = true)
|-- SALARY_TO: integer (nullable = true)
|-- SALARY_FROM: integer (nullable = true)
|-- LOCATION: string (nullable = true)
|-- CITY: string (nullable = true)
|-- CITY_NAME: string (nullable = true)
|-- COUNTY: integer (nullable = true)
|-- COUNTY_NAME: string (nullable = true)
|-- MSA: integer (nullable = true)
|-- MSA_NAME: string (nullable = true)

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|-- STATE: integer (nullable = true)
|-- STATE_NAME: string (nullable = true)
|-- COUNTY_OUTGOING: integer (nullable = true)
|-- COUNTY_NAME_OUTGOING: string (nullable = true)
|-- COUNTY_INCOMING: integer (nullable = true)
|-- COUNTY_NAME_INCOMING: string (nullable = true)
|-- MSA_OUTGOING: integer (nullable = true)
|-- MSA_NAME_OUTGOING: string (nullable = true)
|-- MSA_INCOMING: integer (nullable = true)
|-- MSA_NAME_INCOMING: string (nullable = true)
|-- NAICS2: integer (nullable = true)
|-- NAICS2_NAME: string (nullable = true)
|-- NAICS3: integer (nullable = true)
|-- NAICS3_NAME: string (nullable = true)
|-- NAICS4: integer (nullable = true)
|-- NAICS4_NAME: string (nullable = true)
|-- NAICS5: integer (nullable = true)
|-- NAICS5_NAME: string (nullable = true)
|-- NAICS6: integer (nullable = true)
|-- NAICS6_NAME: string (nullable = true)
|-- TITLE: string (nullable = true)
|-- TITLE_NAME: string (nullable = true)
|-- TITLE_CLEAN: string (nullable = true)
|-- SKILLS: string (nullable = true)
|-- SKILLS_NAME: string (nullable = true)
|-- SPECIALIZED_SKILLS: string (nullable = true)
|-- SPECIALIZED_SKILLS_NAME: string (nullable = true)
|-- CERTIFICATIONS: string (nullable = true)
|-- CERTIFICATIONS_NAME: string (nullable = true)
|-- COMMON_SKILLS: string (nullable = true)
|-- COMMON_SKILLS_NAME: string (nullable = true)
|-- SOFTWARE_SKILLS: string (nullable = true)
|-- SOFTWARE_SKILLS_NAME: string (nullable = true)
|-- ONET: string (nullable = true)
|-- ONET_NAME: string (nullable = true)
|-- ONET_2019: string (nullable = true)
|-- ONET_2019_NAME: string (nullable = true)
|-- CIP6: string (nullable = true)
|-- CIP6_NAME: string (nullable = true)
|-- CIP4: string (nullable = true)
|-- CIP4_NAME: string (nullable = true)
|-- CIP2: string (nullable = true)
|-- CIP2_NAME: string (nullable = true)

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|-- SOC_2021_2: string (nullable = true)
|-- SOC_2021_2_NAME: string (nullable = true)
|-- SOC_2021_3: string (nullable = true)
|-- SOC_2021_3_NAME: string (nullable = true)
|-- SOC_2021_4: string (nullable = true)
|-- SOC_2021_4_NAME: string (nullable = true)
|-- SOC_2021_5: string (nullable = true)
|-- SOC_2021_5_NAME: string (nullable = true)
|-- LOT_CAREER_AREA: integer (nullable = true)
|-- LOT_CAREER_AREA_NAME: string (nullable = true)
|-- LOT_OCCUPATION: integer (nullable = true)
|-- LOT_OCCUPATION_NAME: string (nullable = true)
|-- LOT_SPECIALIZED_OCCUPATION: integer (nullable = true)
|-- LOT_SPECIALIZED_OCCUPATION_NAME: string (nullable = true)
|-- LOT_OCCUPATION_GROUP: integer (nullable = true)
|-- LOT_OCCUPATION_GROUP_NAME: string (nullable = true)
|-- LOT_V6_SPECIALIZED_OCCUPATION: integer (nullable = true)
|-- LOT_V6_SPECIALIZED_OCCUPATION_NAME: string (nullable = true)
|-- LOT_V6_OCCUPATION: integer (nullable = true)
|-- LOT_V6_OCCUPATION_NAME: string (nullable = true)
|-- LOT_V6_OCCUPATION_GROUP: integer (nullable = true)
|-- LOT_V6_OCCUPATION_GROUP_NAME: string (nullable = true)
|-- LOT_V6_CAREER_AREA: integer (nullable = true)
|-- LOT_V6_CAREER_AREA_NAME: string (nullable = true)
|-- SOC_2: string (nullable = true)
|-- SOC_2_NAME: string (nullable = true)
|-- SOC_3: string (nullable = true)
|-- SOC_3_NAME: string (nullable = true)
|-- SOC_4: string (nullable = true)
|-- SOC_4_NAME: string (nullable = true)
|-- SOC_5: string (nullable = true)
|-- SOC_5_NAME: string (nullable = true)
|-- LIGHTCAST_SECTORS: string (nullable = true)
|-- LIGHTCAST_SECTORS_NAME: string (nullable = true)
|-- NAICS_2022_2: integer (nullable = true)
|-- NAICS_2022_2_NAME: string (nullable = true)
|-- NAICS_2022_3: integer (nullable = true)
|-- NAICS_2022_3_NAME: string (nullable = true)
|-- NAICS_2022_4: integer (nullable = true)
|-- NAICS_2022_4_NAME: string (nullable = true)
|-- NAICS_2022_5: integer (nullable = true)
|-- NAICS_2022_5_NAME: string (nullable = true)
|-- NAICS_2022_6: integer (nullable = true)

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|-- NAICS_2022_6_NAME: string (nullable = true)
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|1f57d95acf4dc67ed...|          9/6/2024|  2024-09-06 20:32:...|          0|6/2/2024| 6/8/2024
May-2024\n\nEn...|          6/8/2024|          6|  894731|          Murphy USA| Murphy USA
time (> 32 h...|          2|          2|          false| NULL|          0|
2051.01|Business Intellig...|15-2051.01|Business Intellig...|[\n  "45.0601",\n...|[\n  "Econ
0000|Computer and Math...|  15-2000|Mathematical Scie...|  15-2050|Data Scientists|  15-
2051|Data Scientists|          23|Information Techn...|          231010|Business Intellig..
0000|Computer and Math...|15-2000|Mathematical Scie...|15-2050|Data Scientists|15-
2051|Data Scientists|          [\n  7\n]|  [\n  "Artificial ...|          44|          Retail Tra
|0cb072af26757b6c4...|          8/2/2024|  2024-08-02 17:08:...|          0|6/2/2024| 8/1/2024
time (> 32 h...|          3|          3|          false| NULL|          1|
Watervill...|  23|          Maine|          23011|          Kennebec, ME|          23011|          K
Watervill...|          12300|Augusta-Watervill...|          56|Administrative an...|          561|Administra
2051.01|Business Intellig...|15-2051.01|Business Intellig...|          []|
0000|Computer and Math...|  15-2000|Mathematical Scie...|  15-2050|Data Scientists|  15-
2051|Data Scientists|          23|Information Techn...|          231010|Business Intellig..
0000|Computer and Math...|15-2000|Mathematical Scie...|15-2050|Data Scientists|15-
2051|Data Scientists|          NULL|          NULL|          56|Administrative an
|85318b12b3331fa49...|          9/6/2024|  2024-09-06 20:32:...|          1|6/2/2024| 7/7/2024
time (> 32 h...|          5|          NULL|          false| NULL|          0|
Fort Worth...|  48|          Texas|          48113|          Dallas, TX|          48113|
Fort Worth...|          19100|Dallas-Fort Worth...|          52|Finance and Insur...|          524|Insurance
2051.01|Business Intellig...|15-2051.01|Business Intellig...|          []|

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```

only showing top 5 rows

```

from pyspark.sql.functions import monotonically_increasing_id

companies_df = df.select(
    "COMPANY",
    "COMPANY_NAME",
    "COMPANY_RAW",
    "COMPANY_IS_STAFFING"
).distinct()

# Create primary key
companies_df = companies_df.withColumn(
    "COMPANY_ID",
    monotonically_increasing_id()
)

# Reorder columns
companies_df = companies_df.select(
    "COMPANY_ID",
    "COMPANY",
    "COMPANY_NAME",
    "COMPANY_RAW",
    "COMPANY_IS_STAFFING"
)

# Register SQL table
companies_df.createOrReplaceTempView("companies")

companies_df.show(5)

```

[Stage 3:>

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```

+-----+-----+-----+-----+-----+
+
|COMPANY_ID| COMPANY|          COMPANY_NAME|          COMPANY_RAW|COMPANY_IS_STAFFING|

```



```

+-----+-----+-----+-----+-----+
+
|          0|35247729|          Crowe|          Crowe|          false|
|          1|39461288|The Devereux Foun...|The Devereux Foun...|          false|
|          2|40008275|          Elder Research|          Elder Research|          false|
|          3|37060425|          NTT DATA|          NTT DATA Inc|          false|
|          4|40882284|Frederick Nationa...|Frederick Nationa...|          false|
+-----+-----+-----+-----+-----+
+
only showing top 5 rows

```

```

industries_df = df.select(
    "NAICS_2022_6",
    "NAICS_2022_6_NAME",
    "SOC_5",
    "SOC_5_NAME",
    "LOT_SPECIALIZED_OCCUPATION_NAME",
    "LOT_OCCUPATION_GROUP"
).distinct()

industries_df = industries_df.withColumn(
    "INDUSTRY_ID",
    monotonically_increasing_id()
)

industries_df = industries_df.select(
    "INDUSTRY_ID",
    "NAICS_2022_6",
    "NAICS_2022_6_NAME",
    "SOC_5",
    "SOC_5_NAME",
    "LOT_SPECIALIZED_OCCUPATION_NAME",
    "LOT_OCCUPATION_GROUP"
)

industries_df.createOrReplaceTempView("industries")

industries_df.show(5)

```

[Stage 6:>

(0 + 1) / 1]

INDUSTRY_ID	NAICS_2022_6	NAICS_2022_6_NAME	SOC_5	SOC_5_NAME	LOT_SPECIALIZED_OCCUPATION
0	424490	Other Grocery and...	15-2051	Data Scientists	SAP Analysts
1	562111	Solid Waste Collec...	15-2051	Data Scientists	Oracle Consultants
2	457110	Gasoline Stations...	15-2051	Data Scientists	Oracle Consultants
3	334111	Electronic Comput...	15-2051	Data Scientists	Oracle Consultants
4	621320	Offices of Optome...	15-2051	Data Scientists	General ERPs

only showing top 5 rows

```
locations_df = df.select(
    "LOCATION",
    "CITY_NAME",
    "STATE_NAME",
    "COUNTY_NAME",
    "MSA",
    "MSA_NAME"
).distinct()

locations_df = locations_df.withColumn(
    "LOCATION_ID",
    monotonically_increasing_id()
)

locations_df = locations_df.select(
    "LOCATION_ID",
    "LOCATION",
    "CITY_NAME",
    "STATE_NAME",
    "COUNTY_NAME",
    "MSA",
    "MSA_NAME"
)

locations_df.createOrReplaceTempView("locations")
```

```
locations_df.show(5)
```

[Stage 9:>

(0 + 1) / 1]

```
+-----+-----+-----+-----+-----+
+-----+-----+
|LOCATION_ID|          LOCATION|          CITY_NAME|STATE_NAME|COUNTY_NAME|  MSA|
+-----+-----+-----+-----+-----+
+-----+-----+
|          0|{\n  "lat": 24.55...|          Key West, FL|    Florida| Monroe, FL|28580|          Key
|          1|{\n  "lat": 41.62...|          Urbandale, IA|        Iowa|   Polk, IA|19780|Des Moines-
West D...|
|          2|{\n  "lat": 37.20...|    Springfield, MO|    Missouri| Greene, MO|44180|    Spring
|          3|{\n  "lat": 40.74...|[Unknown City], NJ|New Jersey| Hudson, NJ|35620|New York-
Newark-J...|
|          4|{\n  "lat": 40.79...|    West Orange, NJ|New Jersey|  Essex, NJ|35620|New York-
Newark-J...|
+-----+-----+-----+-----+-----+
+-----+-----+
only showing top 5 rows
```

```
job_postings_df = df.select(
    "ID",
    "TITLE_CLEAN",
    "EMPLOYMENT_TYPE_NAME",
    "REMOTE_TYPE_NAME",
    "BODY",
    "MIN_YEARS_EXPERIENCE",
    "MAX_YEARS_EXPERIENCE",
    "SALARY",
    "SALARY_FROM",
    "SALARY_TO",
    "POSTED",
    "EXPIRED",
    "DURATION",
    "COMPANY",
    "LOCATION",
    "NAICS_2022_6"
)
```

```

job_postings_df = job_postings_df.join(
    companies_df.select("COMPANY_ID", "COMPANY"),
    on="COMPANY",
    how="left"
)

job_postings_df = job_postings_df.join(
    locations_df.select("LOCATION_ID", "LOCATION"),
    on="LOCATION",
    how="left"
)

job_postings_df = job_postings_df.join(
    industries_df.select("INDUSTRY_ID", "NAICS_2022_6"),
    on="NAICS_2022_6",
    how="left"
)

```

```

job_postings_df = job_postings_df.select(
    "ID",
    "TITLE_CLEAN",
    "COMPANY_ID",
    "INDUSTRY_ID",
    "EMPLOYMENT_TYPE_NAME",
    "REMOTE_TYPE_NAME",
    "BODY",
    "MIN_YEARS_EXPERIENCE",
    "MAX_YEARS_EXPERIENCE",
    "SALARY",
    "SALARY_FROM",
    "SALARY_TO",
    "LOCATION_ID",
    "POSTED",
    "EXPIRED",
    "DURATION"
)

job_postings_df.createOrReplaceTempView("job_postings")

job_postings_df.show(5)

```

ID	TITLE_CLEAN	COMPANY_ID	INDUSTRY_ID	EMPLOYMENT_TYPE_NAME	REMOTE	
1f57d95acf4dc67ed...	enterprise analys...	14586	1753	Full-		
time (> 32 h...	[None] 31-May-2024\n\nEn...		2			
1f57d95acf4dc67ed...	enterprise analys...	14586	1616	Full-		
time (> 32 h...	[None] 31-May-2024\n\nEn...		2			
1f57d95acf4dc67ed...	enterprise analys...	14586	1024	Full-		
time (> 32 h...	[None] 31-May-2024\n\nEn...		2			
1f57d95acf4dc67ed...	enterprise analys...	14586	942	Full-		
time (> 32 h...	[None] 31-May-2024\n\nEn...		2			
1f57d95acf4dc67ed...	enterprise analys...	14586	400	Full-		
time (> 32 h...	[None] 31-May-2024\n\nEn...		2			

only showing top 5 rows

```

query1 = """
SELECT
    i.NAICS_2022_6_NAME AS industry_name,
    i.LOT_SPECIALIZED_OCCUPATION_NAME AS specialized_occupation,
    PERCENTILE_APPROX(j.SALARY, 0.5) AS median_salary
FROM job_postings j
JOIN industries i
    ON j.INDUSTRY_ID = i.INDUSTRY_ID
WHERE
    i.NAICS_2022_6 = '518210'
    AND j.SALARY IS NOT NULL
    AND j.SALARY > 0
GROUP BY
    i.NAICS_2022_6_NAME,
    i.LOT_SPECIALIZED_OCCUPATION_NAME
ORDER BY
    median_salary DESC
"""

```

```
query1_result = spark.sql(query1)
```

```
query1_result.show(20, truncate=False)
```

[Stage 34:>

(0 + 1) / 1]

```
+-----+-----+
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|industry_name|specialized_occupation|
+-----+-----+
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Financial Services|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Data Processing|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Marketing|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Data Processing|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Enterprise Software|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|General Business|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Business Services|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Oracle|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|SAP|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Business Services|
|Computing Infrastructure Providers, Data Processing, Web Hosting, and Related Services|Healthcare|
+-----+-----+
+-----+-----+
```

```
query1_pdf = query1_result.toPandas()
```

```
import plotly.express as px

fig = px.bar(
    query1_pdf,
    x="median_salary",
    y="specialized_occupation",
    color="industry_name",
    orientation="h",
```

```

        title="Median Salary by Specialized Occupation in Technology Industry"
    )

fig.show()

```

Unable to display output for mime type(s): application/vnd.plotly.v1+json

The median salaries of each specialized occupation in the technology industry are very close to one another, thus suggesting no matter the profession, this industry is a profitable one to work in for salaries of around 120k-130k

Query 2 — Top 5 Companies with the Most Remote Jobs in California

```

query2 = """
SELECT
    c.COMPANY_NAME AS company_name,
    COUNT(*) AS remote_jobs
FROM job_postings j
JOIN companies c
    ON j.COMPANY_ID = c.COMPANY_ID
JOIN locations l
    ON j.LOCATION_ID = l.LOCATION_ID
WHERE
    j.REMOTE_TYPE_NAME = 'Remote'
    AND l.STATE_NAME = 'California'
GROUP BY
    c.COMPANY_NAME
ORDER BY
    remote_jobs DESC
LIMIT 5
"""

```

```
query2_result = spark.sql(query2)
```

```
query2_result.show(truncate=False)
```

[Stage 66:>

(0 + 1) / 1]

company_name	remote_jobs
Unclassified	1509354
University of California	9438
Kaiser Permanente	2541
Amentum	1650
Randstad	1485

```
query2_pdf = query2_result.toPandas()
```

```
import seaborn as sns
import matplotlib.pyplot as plt

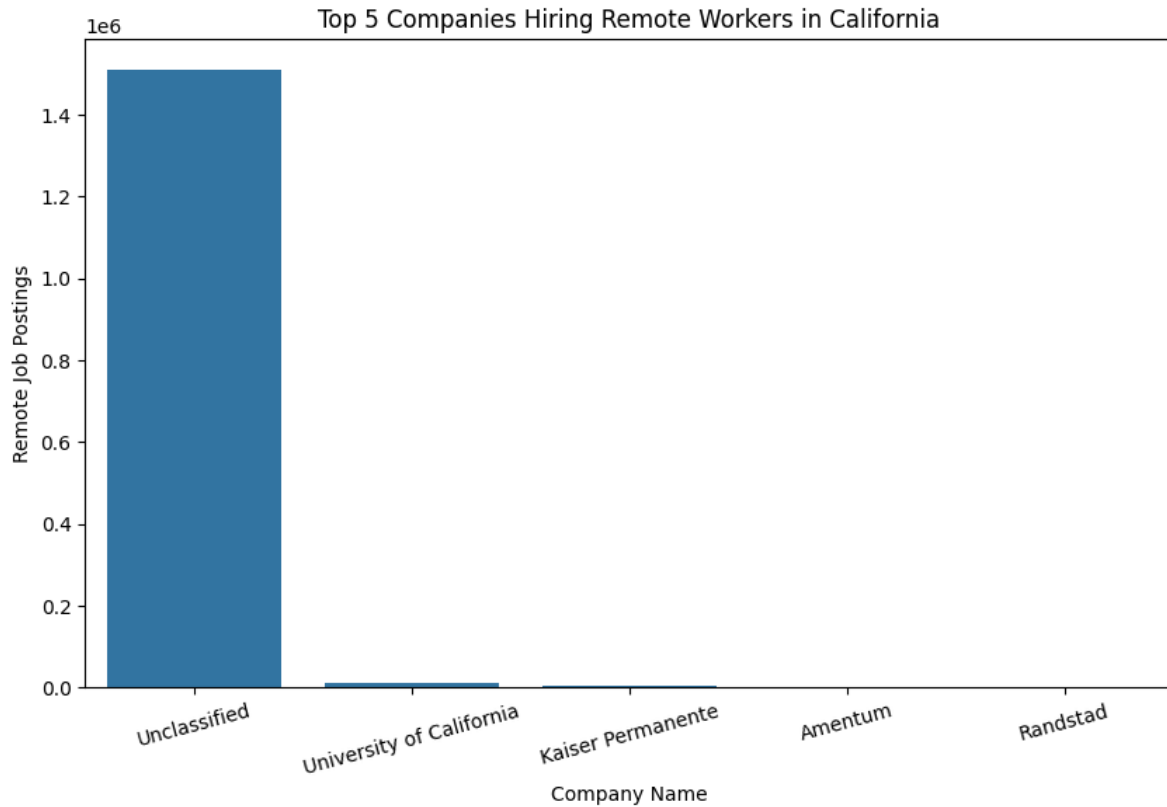
plt.figure(figsize=(10,6))

sns.barplot(
    data=query2_pdf,
    x="company_name",
    y="remote_jobs"
)

plt.title("Top 5 Companies Hiring Remote Workers in California")
plt.xlabel("Company Name")
plt.ylabel("Remote Job Postings")

plt.xticks(rotation=15)

plt.show()
```

Though it may be skewed heavily towards unclassified, the number of remote opportunities at some of the largest hiring firms in the industry not having a very significant comparison to the smaller firms grouped together does give insight to how smaller firms likely utilize remote positions more than larger industry leaders

Query 3 — Monthly Job Posting Trends in California

```
query3 = """
SELECT
    YEAR(try_to_timestamp(j.POSTED, 'M/d/yyyy')) AS year,
    MONTH(try_to_timestamp(j.POSTED, 'M/d/yyyy')) AS month,
    COUNT(*) AS job_count
FROM job_postings j
JOIN locations l
    ON j.LOCATION_ID = l.LOCATION_ID
WHERE
    l.STATE_NAME = 'California'
```

```

        AND try_to_timestamp(j.POSTED, 'M/d/yyyy') IS NOT NULL
GROUP BY
    YEAR(try_to_timestamp(j.POSTED, 'M/d/yyyy')),
    MONTH(try_to_timestamp(j.POSTED, 'M/d/yyyy'))
ORDER BY
    year DESC,
    month DESC
"""

```

```
query3_result = spark.sql(query3)
```

```
query3_result.show(20, truncate=False)
```

[Stage 115:>

(0 + 1) / 1]

```

+----+-----+-----+
|year|month|job_count|
+----+-----+-----+
|2024|9    |2787841  |
|2024|8    |1834724  |
|2024|7    |1787680  |
|2024|6    |1954292  |
|2024|5    |1658115  |
+----+-----+-----+

```

```
query3_pdf = query3_result.toPandas()
```

```

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))

sns.lineplot(
    data=query3_pdf,
    x="month",

```

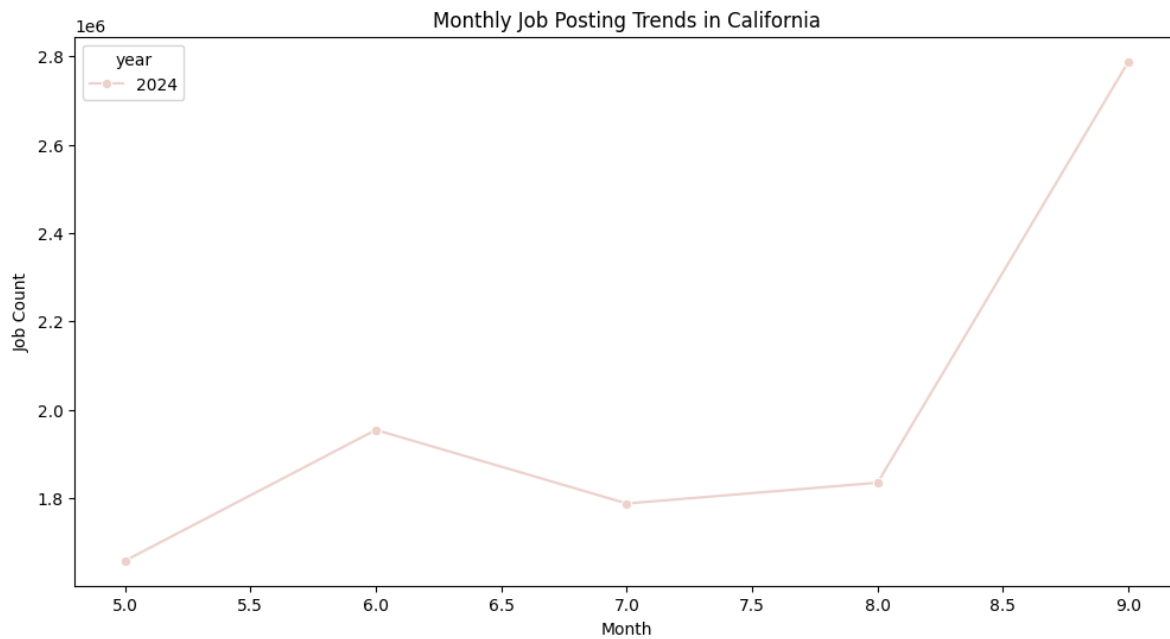
```

    y="job_count",
    hue="year",
    marker="o"
)

plt.title("Monthly Job Posting Trends in California")
plt.xlabel("Month")
plt.ylabel("Job Count")

plt.show()

```



As the months go on, the number of job opportunities increases overall, with a significant rise in month 8 onward and slight dip between month 6 and 7. This showcases the economic patterns of the industry, as with more job counts in the later months, there likely is more economic activity and thus justification to hire more workers.

Query 4 — Salary Comparisons Across Major US Cities

```

query4 = """
SELECT
CASE

```

```

        WHEN l.MSA = '14460' THEN 'Boston'
        WHEN l.MSA = '47900' THEN 'Washington DC'
        WHEN l.MSA = '35620' THEN 'New York'
        WHEN l.MSA = '41860' THEN 'San Francisco'
        WHEN l.MSA = '42660' THEN 'Seattle'
        WHEN l.MSA = '31080' THEN 'Los Angeles'
        WHEN l.MSA = '19100' THEN 'Dallas'
        WHEN l.MSA = '26420' THEN 'Houston'
        WHEN l.MSA = '12420' THEN 'Austin'
        WHEN l.MSA = '34980' THEN 'Nashville'
        WHEN l.MSA = '28140' THEN 'Kansas City'
        WHEN l.MSA = '19740' THEN 'Denver'
    END AS metro_area,

    ROUND(AVG(j.SALARY), 2) AS avg_salary,

    COUNT(*) AS job_count

FROM job_postings j

JOIN locations l
    ON j.LOCATION_ID = l.LOCATION_ID

WHERE
    j.SALARY IS NOT NULL
    AND j.SALARY > 0
    AND l.MSA IN (
        '14460',
        '47900',
        '35620',
        '41860',
        '42660',
        '31080',
        '19100',
        '26420',
        '12420',
        '34980',
        '28140',
        '19740'
    )

GROUP BY

```

```
1.MSA

ORDER BY
    avg_salary DESC
"""
```

```
query4_result = spark.sql(query4)
```

```
query4_result.show(truncate=False)
```

```
[Stage 149:=====>
```

```
(1 + 1) / 2]
```

```
+-----+-----+-----+
|metro_area|avg_salary|job_count|
+-----+-----+-----+
|Washington DC|135606.59|2287790|
|San Francisco|133712.35|1336910|
|Seattle      |127412.69|655292|
|Houston      |116758.83|1402474|
|Los Angeles  |115888.81|1700450|
|New York     |111576.76|2143319|
|Boston       |111506.67|690133|
|Nashville    |110872.23|101198|
|Austin       |110532.12|2686063|
|Dallas       |106088.88|2264019|
|Denver       |105159.8|1686604|
|Kansas City  |77262.74|169339|
+-----+-----+-----+
```

```
query4_pdf = query4_result.toPandas()
```

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
```

```

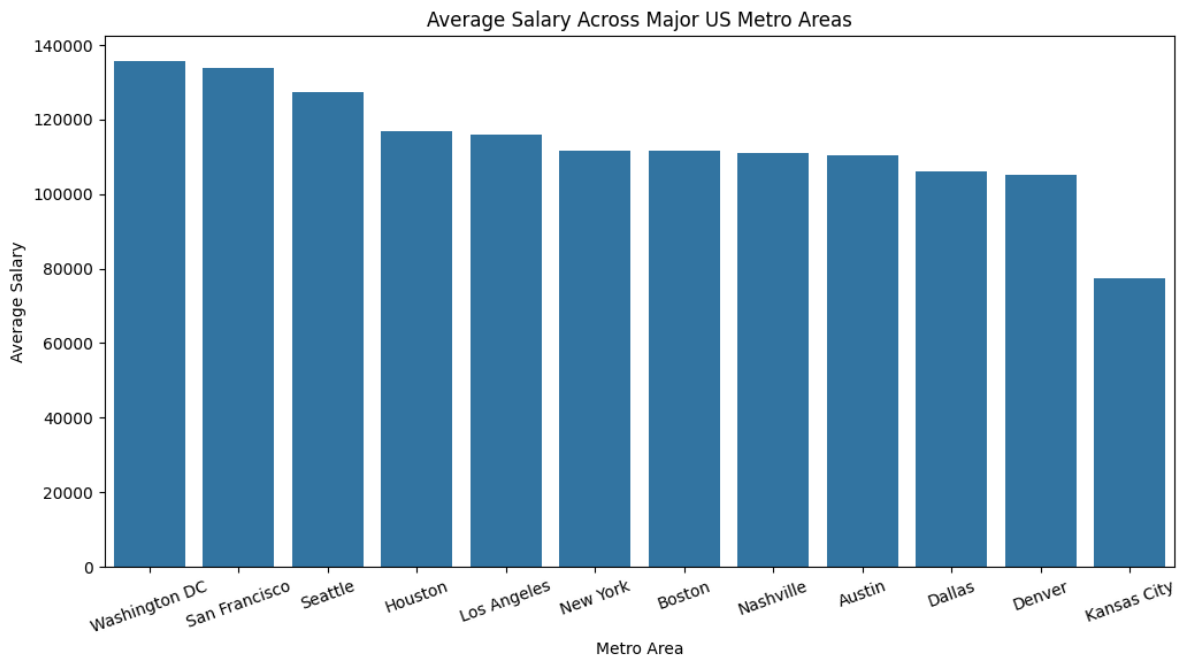
sns.barplot(
    data=query4_pdf,
    x="metro_area",
    y="avg_salary"
)

plt.title("Average Salary Across Major US Metro Areas")
plt.xlabel("Metro Area")
plt.ylabel("Average Salary")

plt.xticks(rotation=20)

plt.show()

```



Again, most of the metro areas are relatively close to one another, however, it is clear that the predominantly east and west coast areas have higher average salaries than otherwise. Houston is the one exception until Nashville of a non-east or non-west coast metro area being in the top 7, suggesting greater economic activity and opportunity in these regions.